RescueSwarm: An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

Technical Design Proposal

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***Abstract*—India recorded 5.4 million internal displacements from disasters in 2024, the highest figure in twelve years, and the binding constraint on survivor outcomes is the speed with which casualties are located and reached. Ground and boat-based search is slow and hazardous; helicopter search is fast but scarce and expensive. This proposal describes RescueSwarm, a three-aircraft autonomous unmanned aerial system that surveys a designated area, detects and geolocates survivors, and delivers a standardised relief kit to each, under a single operator and without any reliance on cellular or internet infrastructure. Components are sourced domestically where a qualifying part exists, principally because domestic lead times are roughly half those of imports and therefore buy flight-test days inside a schedule-limited programme; we state which lines were deferred and what each omission costs the system rather than the budget. We present the system architecture, the sizing and error models that fix its design point, the simulation evidence obtained to date, and a phased work plan. We are explicit throughout about which figures are measured, which are modelled, and which remain assumed; no aircraft has yet flown, and the proposal is structured so that a reviewer can distinguish demonstrated capability from projected capability without reading the source material.**

***Index Terms*—unmanned aerial systems, multi-robot coordination, search and rescue, coverage path planning, RTK geolocation, edge inference, disaster response**

I. Introduction and Problem Statement

A. The operational gap

India records among the highest disaster-displacement figures in the world. The Internal Displacement Monitoring Centre reports **5.4 million internal displacements in India during 2024** — the highest in twelve years — of which 2.4 million were triggered by a single monsoon flood season in Assam [1]. The scale of the response problem is therefore not in question; what is in question is how quickly the people displaced by it can be found.

Post-flood search and rescue in India is dominated by a single quantity: the elapsed time between the onset of the event and the moment a survivor is both *located* and *reached*. Nationally deployed capability addresses the second of these far better than the first. Boat teams and ground parties can reach a survivor once told where to go, but they search slowly, at high risk to the crew, and with poor coverage of debris fields and rooftops. Rotary-wing aircraft search quickly but are scarce, costly per flight hour, and are typically committed to evacuation rather than to systematic area survey.

The consequence is that *localisation* is the bottleneck. A search asset that can systematically cover an assigned polygon, detect human figures in cluttered post-flood imagery, and report each detection as a coordinate accurate to a few metres would materially change the tasking of every downstream resource.

B. Why a swarm rather than a single aircraft

A single multirotor with a useful payload and a fifteen-minute endurance covers a disappointing area. Coverage scales with the number of simultaneously operating aircraft, but so does operator workload, and a system requiring one pilot per aircraft does not deploy well in a disaster. The engineering problem is therefore not simply flight, but *coordinated autonomous flight under a single operator*: area decomposition, task allocation, deconfliction, consolidated reporting, and graceful degradation when the command link is lost.

C. Why sourcing is a schedule decision

Domestic sourcing is treated here as a schedule constraint rather than a policy position. Our supplier survey puts typical import lead times at four to six weeks against two to three weeks domestically. In a programme whose binding constraint is an eight-week flight-test window, that difference is not a procurement detail: it is the difference between two test campaigns and one. Suppliers are therefore named specifically throughout, and lead time is carried alongside price wherever a part is selected.

This argument cuts against a decision taken later in the document. The build defers the Indian autopilot and Indian propulsion on cost, and the imported parts replacing them carry the longer lead time just described as decisive. The mitigation is structural: every long-lead item is ordered in tranche 1, in month one, ahead of the software work that would normally precede procurement. Cost was traded against schedule margin deliberately, and the resulting schedule risk is carried in Table IV.

D. Contributions of the proposed work

1) A three-aircraft autonomous survey-and-delivery system operable by a single operator with no network dependency, sized line-by-line against market-verified components.
2) A geolocation error budget that treats survivor coordinate accuracy as the primary system output, and an architecture that meets it using a team-owned RTK base rather than any external correction service.
3) A coordination scheme — static equal-area decomposition with boustrophedon coverage and greedy claim-and-lock task allocation — selected for determinism and verifiability over optimality.
4) An evidence discipline in which every published number regenerates from its model in continuous integration, and in which measured, modelled and assumed quantities are separated by construction.

II. Objectives and Scope

A. Primary objectives

O1 Survey a designated area autonomously with three cooperating aircraft under one operator and one ground control station.

O2 Detect human figures in nadir imagery at a ground sample distance sufficient for reliable classification, with onboard inference and multi-frame temporal fusion.

O3 Report each detection as a geodetic coordinate with a circular error probable within a few metres.

O4 Deliver a standardised relief kit to each confirmed survivor location by controlled free-fall release.

O5 Operate with no cellular, internet or cloud dependency at any point in the mission.

B. Explicit non-objectives

These are stated because proposals of this kind are frequently over-scoped, and because each exclusion is a deliberate engineering decision rather than an omission.

- **Not a casualty-evacuation platform.** The payload class is a relief kit, not a person, and no growth path to the latter is claimed.
- **Not beyond-visual-line-of-sight certified.** The design operates within a bounded geofence; regulatory expansion is future work.
- **Not a night or all-weather system.** The perception stack is visible-band. Night capability is future work; Section XI argues that near-infrared rather than thermal is the extension matched to a target immersed in water.
- **Not a survivor-triage system.** The system reports position and delivers a kit; it makes no medical assessment.

III. Related Work and State of the Art

A. Coverage path planning

Area decomposition for multi-robot coverage is mature. Divide Areas for Optimal Multi-Robot Coverage (DARP) produces equal-area, connected, robot-specific regions and is the natural reference method [2]. Classical boustrophedon decomposition [3] remains the standard in-region motion primitive for sensor footprints that are rectangular in the body frame.

We adopt a static equal-area partition with boustrophedon in-region coverage rather than a full DARP implementation. This is a deliberate trade of optimality for determinism: a static partition produces an identical, inspectable plan for identical inputs, which is a precondition for the verification regime described in Section VI-B. DARP is retained as design rationale and as a future-work path where dynamic reallocation becomes necessary.

B. Multi-robot task allocation

The Consensus-Based Bundle Algorithm (CBBA) is the canonical decentralised auction method for multi-agent assignment [4]. Its guarantees depend on a consensus phase over a communication graph. For a three-aircraft system with a reliable star topology to a single ground station, we implement greedy claim-and-lock with a deterministic tie-break, which is materially simpler to verify and whose worst-case behaviour on three agents is well within acceptable bounds.

C. Aerial detection of people

Small-object detection from altitude is the governing difficulty: a supine adult at survey altitude occupies a few tens of pixels. Slicing-aided hyper inference (SAHI) and related tiling strategies substantially improve small-object recall by running detection on overlapping crops rather than a downsampled full frame [5]. The difficulty is quantified by the size bands used in the standard benchmarks: COCO classifies an object below $32^2\,\mathrm{px}^2$ as small [6], and TinyPerson introduces finer *tiny* bands specifically because aerial person detection falls below even that threshold [7].

Purpose-built aerial person datasets provide the transfer-learning base. SARD [8] covers land search and rescue and the broader VisDrone benchmark [9] covers urban aerial scenes, but the closest match to the target condition here is SeaDronesSee [10], which annotates people in open water from a UAV and therefore shares both the low-contrast background and the head-and-shoulders target signature of a flood scene. None contains post-flood Indian terrain. A field data campaign is therefore the right answer; it is *deferred* from this build (Section VIII), which is why detection recall remains a modelled quantity and the leading entry in the risk register.

TABLE I
AIR VEHICLE DESIGN POINT (PER AIRCRAFT)

Parameter	**Value**
Configuration	Quadrotor
Rotor diameter	18 in
Maximum take-off mass	6.36 kg
Fleet mass (3 aircraft)	19.08 kg
Regulatory fleet cap	25 kg [13]
Battery	6S3P Li-ion 21700, 292 Wh
Pack nominal voltage	21.6 V (18.0–25.2 V)
Pack mass	1 449 g
Cells per aircraft	18
Peak pack current	115 A (8.5 C) at T/W 2.0
Static thrust-to-weight	2.0
Hover endurance	15.3 min
Design mission duration	7.6 min
Survey altitude	40 m AGL
Ground sample distance	$\leq$2.0 cm/px
Survey groundspeed	8 m/s
Payload	4 kits $\times$ 200 g
Geofence radius	600 m

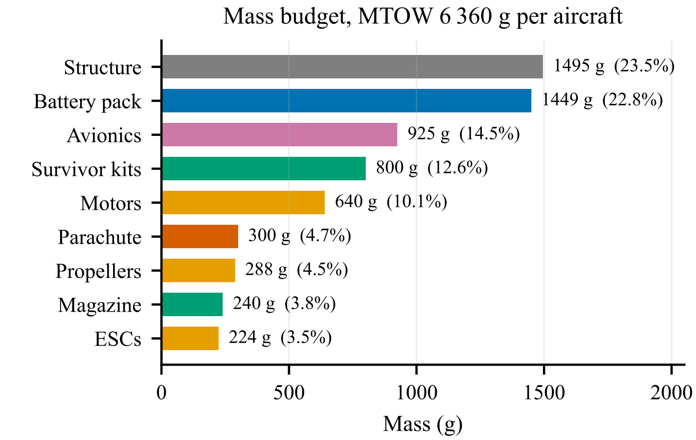

Fig. 1. Mass budget per aircraft. Generated from `docs/sizing/model-output.txt`. *Avionics* includes the wiring harness and *magazine* the release mechanism. The fleet figure of 19 081 g carries 4 919 g of growth allowance against a 24 kg internal target — 26 % build overweight tolerated — inside the 25 kg regulatory cap.

D. Airborne payload delivery

The dominant error term in airborne delivery is release state — velocity and wind — rather than release altitude. Mathisen *et al.* demonstrate autonomous ballistic airdrop from a small fixed-wing UAV and report a mean delivery error of 5.5 m, with the release state computed dynamically against wind and vehicle velocity and re-optimised in flight precisely because that term dominates [11]. This is the decisive argument for a multirotor in this role: a rotorcraft can null its groundspeed to near zero before release and thereby remove the dominant error term outright, which a fixed-wing cannot. The delivery budget in Section IV-I exploits exactly this.

E. Positioning

Real-time kinematic correction from a locally surveyed base is standard practice and is the only route to sub-metre relative positioning without an external network [12]. Since the system may not use internet or cellular services, a team-owned base station is not merely preferable but required.

IV. PROPOSED SYSTEM ARCHITECTURE

A. Design point

Table I states the aircraft design point. Every value is produced by a sizing model held in the repository and regenerated in continuous integration; none is transcribed by hand.

This is the reference design point, and one line of it is not yet verified for the aircraft actually being built. The adopted configuration (Section VIII) uses generic motors that publish no thrust curve, so the thrust-to-weight figure below is a requirement rather than a measurement until the P5 thrust stand runs. Nothing else in the table moves: mass, energy, altitude, speed and payload are all set by the mission.

The thrust-to-weight ratio of 2.0 and the hover endurance of 15.3 minutes against a 7.7-minute design mission are reserve policies rather than performance targets, because the failure mode that ends a sortie is an aircraft that does not return. The energy reserve is stated precisely in Section V; it, and not the design mission, is what sizes the pack.

Fig. 1 gives the mass statement behind those figures. Two features drive the rest of the design. Structure and battery together are 46 % of maximum take-off mass, which is why airframe mass fraction and pack chemistry are the two sensitivities carried in the risk register rather than, say, avionics choice.

The itemisation closes to maximum take-off mass exactly, and the model asserts that it does. The largest discretionary line is the recovery parachute and its mount at 300 g, 4.7 % of the aircraft and the only mass carried against loss of the airframe rather than against the mission. The survivor kits are 800 g — 12.6 % of MTOW and fixed by the mission at four kits of exactly 200 g — so payload is an input to the sizing loop, not an output of it. This is why propulsion and energy resist cost reduction in a way that avionics does not: they are sized by a payload that is not negotiable, so cheaper parts of the same class are available but a smaller aircraft is not.

B. Energy: the decisions behind the pack

Section V derives the pack requirement in full — the reserve policy that sizes it, the nameplate energy floor, and the continuous current figure that disqualifies most candidate packs. The re-sweep term in that policy exists because a search finding nothing on its first pass is a normal outcome rather than a failure; the loiter term exists because a recovery queue of three aircraft is not instantaneous.

Two things about the result are decisions rather than arithmetic, and they belong here rather than in a derivation.

Why cylindrical Li-ion rather than LiPo. We record this as a trade rather than a preference, because LiPo wins part of it. A LiPo pack of equal energy would roughly halve the internal resistance — about 2.5 V of sag at peak against

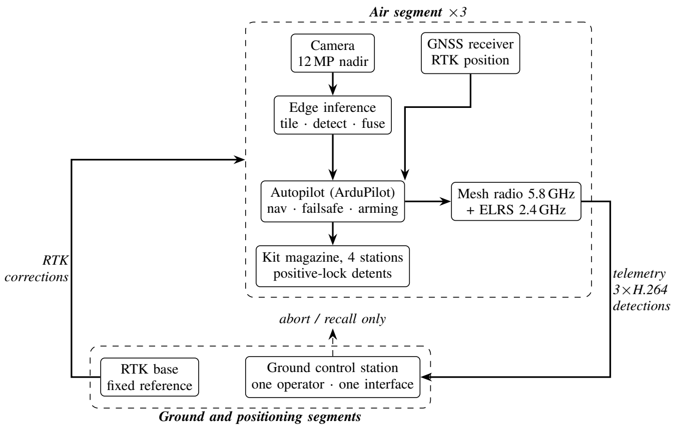

Fig. 2. System architecture. The only permitted downlink commands are abort and recall; the ground station is architecturally incapable of originating a retask, waypoint change or drop command. Detection runs onboard, so the downlink serves the operator rather than the algorithm.

4.6 V — and would carry far more current headroom, and it would remove the pack-assembly step entirely. It loses on three grounds that matter more to this programme: it is approximately 400 g heavier per aircraft at equal energy, which spends fleet margin; its cycle life is two to three times worse across a long flight-test campaign; and it is not domestically sourced, which would remove one of only four remaining Indian lines in Section VIII. The decision is therefore made on campaign cost and sourcing, not on the electrical argument, which points the other way.

Where the reserve is evaluated. A reserve policy is only meaningful at the mass the aircraft actually flies. The pack is rounded up to a buildable 18 cells, which lifts maximum take-off mass roughly 0.7 kg above the value at which the sizing loop converges, so the requirement is re-evaluated at the final design point rather than at the solve. Evaluated there it is about a fifth higher, and the chemistry-agnostic nameplate floor rises with it. The pack meets the policy at the adopted mass, on a margin materially smaller than an evaluation at the solve would report — which is why the check is placed after the loop rather than inside it.

C. Segment architecture

The system comprises three segments, shown in Fig. 2.

1) Air segment: Three identical quadrotors, each carrying a Pixhawk-standard autopilot running ArduPilot [14], a multi-band NavIC/GNSS receiver providing RTK position (Teravolt AeroNav-Pro RTK, tracking NavIC, GPS, GLONASS, Galileo, BeiDou and QZSS on L1/L2/L5/L6 at 25 Hz, specified at 1 cm + 1 ppm in RTK [15]), a nadir-mounted 12 MP camera, an edge inference module, a four-station kit magazine with positive-lock detents, a 5.8 GHz mesh radio for telemetry and video, and a 2.4 GHz ExpressLRS link carrying safety-pilot control and abort independently of the autonomy stack.

2) Ground segment: A single ground control station presenting one unified operator interface. The station is architecturally incapable of originating a retask, waypoint change or drop command; abort and recall are the only permitted downlink commands, and are implemented as an explicit exception. This is a structural safety property rather than a policy: the relevant code paths are absent from the deployed build, not merely guarded.

3) Positioning segment: A team-owned RTK base station, positioned and set to a fixed reference from its first three-dimensional fix within the setup window. Launch is gated on a three-dimensional fix rather than an RTK fix; the first geotag is gated on RTK-fixed, and detections made before convergence are geotagged in float and re-fused once fixed.

What that decoupling costs and buys is now a supplier figure rather than an estimate. The supplier specifies a survey-in of ten to twenty minutes to place the base to ±5 cm [15]. Skipping it removes that entire block from the setup window, which is a far larger saving than the ninety seconds previously claimed here. It is not free: a base set from its own first three-dimensional fix inherits that fix's *absolute* error, of order a metre, and every reported survivor coordinate is offset by the same amount. What RTK preserves is the *relative* accuracy in Table II — the aircraft is centimetric with respect to the base wherever the base happens to think it is.

For this mission that trade is the right way round. A recovery crew navigates to the reported coordinate on its own metre-class receiver, so a shared sub-metre offset on every target is absorbed by the crew's own positioning, whereas twenty minutes of survey-in comes out of the flying window. The survey is worth running whenever the base can be set up before the window opens, and P7–P8 use it, because a measured accuracy figure must be referenced to something better than the system being measured.

A second receiver from the same supplier, the AeroNav-X5, was evaluated against the baseline part and rejected on cost. It is built on a Septentrio GNSS+ core with 448 tracking channels and specifies 0.6 cm + 0.5 ppm horizontal RTK against the baseline part's 1 cm + 1 ppm, with AIM+ interference mitigation. The supplier quotes it at six times the price of the baseline receiver, so the four units this programme needs would consume the greater part of the whole budget. The geolocation budget in Section IV-D is met by the baseline part, and a four-millimetre improvement in a term that is not dominant does not justify that. The baseline receiver is adopted.

Corrections reach the aircraft over the existing command link rather than a dedicated correction radio. This is a documented mode of both the autopilot and the receiver rather than an assumption on our part: ArduPilot's `GPS_INJECT_TO` forwards RTCM 3.x received over telemetry to the receiver's second serial port, which is the format and port the receiver specifies [15]. Removing that radio saves mass, a serial port and a cost line on every aircraft. The same receiver falls back to SBAS when corrections are lost, so the degraded mode needs no additional hardware either.

TABLE II
GEOLOCATION ERROR BY CONFIGURATION

Case	Configuration	RSS error
A	No RTK, single-frame	3.09 m
B	RTK, single-frame	∼1.3 m
C	RTK + multi-frame fusion + calibrated ground plane	**0.91 m**

Case C assumes moving-baseline heading. That receiver is deferred (Section VIII), which substitutes a magnetometer-class attitude term and gives **0.94 m** — an 0.03 m penalty, and the reason the deferral is judged affordable.

D. Geolocation error budget

Survivor coordinate accuracy is the primary system output, and the error budget is therefore the governing analysis of the whole design. Total error is the root sum of squares of GNSS position error, boresight and lever-arm residual, attitude error, terrain-model error and timing skew.

Table II shows the dominant conclusion: positioning quality, not optics, governs. This is what makes RTK a mission requirement rather than an optimisation, and it is why the RTK receiver and its base station survive every cost-reduction pass while a great deal else does not. The *second* receiver, which would supply moving-baseline heading, is a different case and is deferred — see Section VIII.

Multi-frame fusion is effectively free. At the 40 m survey altitude the along-track footprint is 31.4 m, so at 8 m/s groundspeed a target remains in view for 3.9 s. A conservative 2 Hz inference rate therefore yields approximately eight independent observations of each target per pass, and the 3 Hz rate assumed for temporal confirmation yields twelve. Those observations are used both for temporal confirmation, which suppresses false positives, and for geolocation averaging.

Fig. 3 shows both effects quantitatively. Panel (a) makes the positioning argument unarguable: losing the RTK fix moves the error by nearly an order of magnitude, and a 3-D-only fix at 3.67 m RSS consumes most of the 5 m delivery allowance before ballistics or position-hold error are counted. Panel (b) shows why fusion is worth having but not worth over-investing in — it divides the random terms by $\sqrt{n}$ and leaves boresight and target extent untouched, so the curve departs from the ideal $1/\sqrt{n}$ line and flattens by about five frames. Twenty frames is not twice as good as five, and the eight to twelve frames the geometry supplies for free already sit past the knee.

E. Perception and inference budget

The sensor is a 12.3 MP module, 4056×3040 on a **1.55 µm** pitch behind a 6 mm lens, and the frame is tiled at **native resolution** — 48 crops of 640×640 at 20 % overlap, with no downsampling. At the adopted 2 Hz that is **96 inferences per second**, against the 130–160 frames per second this accelerator is measured at for the same model and input size. It fits, with margin.

An earlier revision downsampled by two before tiling, cutting 48 crops to 12. That decision was taken against a different accelerator — one with roughly a quarter of the throughput, and not the part in this bill of materials — and it discarded half the linear resolution the sensor had already delivered. Tile count depends on pixel count alone, so it does not move with altitude; only the target size does.

What the detector is handed does move with altitude, and the effect is larger than it looks:

AGL	Tiling	Target (px)	Area (px^2)	COCO class
40 m	**native**	**39**	**1 498**	medium
60 m	native	26	666	**small**
40 m	2× downsample	19	375	**small**
60 m	2× downsample	13	166	**small**

The target here is a person *in water* — head and shoulders, about 0.4 m across — which is the worst posture and therefore the one the design is sized against. A supine adult on a rooftop is four times easier.

The size class, not the pixel count, is the number that matters. Published recall figures report average precision *by* size band, and small-object AP is routinely far below medium on the same model. Only the top row clears the COCO 32^2 px^2 small-object threshold — which is why the design point is 40 m and native tiling, and why the two changes had to be made together. Either alone leaves the target in the band where detectors do worst.

Two modelling choices set these figures. The optics are computed for the 1.55 µm sensor the bill of materials buys, not for a generic sensor of the same pixel *count* at a different pixel *size*: it is pitch, not count, that fixes ground sample distance. And the target is taken as a person in water, presenting head and shoulders, rather than a supine adult — a fourfold difference in apparent area, and the conservative choice for a flood mission.

Together these fix the design point. Finer ground sampling narrows the field, so coverage costs more transects; but at 40 m those transects cost seconds of a mission with minutes to spare, and the mission is in fact *shorter* than at 60 m because the climb is lower.

This is the single most benchmark-sensitive figure in the design. Published throughput for comparable models on comparable hardware spans roughly 24 to 41 frames per second, and the lower end of that range leaves no margin. Hardware benchmarking before design freeze is accordingly a funded, scheduled activity rather than an assumption.

F. Why the compute stack is three parts

The air vehicle carries a sensor, a single-board host and a neural accelerator. On an aircraft this size that invites the question of why it is not one part, and the answer is that each does something the others cannot.

The accelerator is an **M.2 module whose only interface is PCIe**. It has no camera input, no image signal processor and no general-purpose cores; it executes a compiled graph on tensors a host hands it. The sensor emits a raw Bayer-mosaic stream,

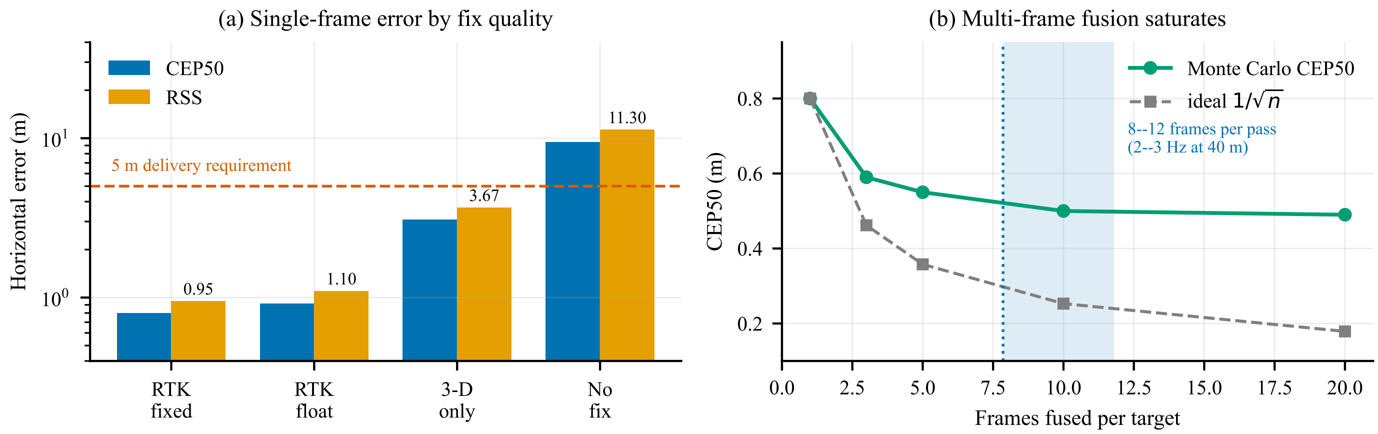

Fig. 3. Geolocation error, from a Monte Carlo run through the same projection code the aircraft flies. (a) Single-frame horizontal error against GNSS fix quality, typical attitude, on a logarithmic scale; the dashed line is the 5 m delivery requirement. (b) Effect of fusing multiple observations of the same target, RTK fixed. Generated by `figures/make_figures.py` from `docs/sizing/geotag-accuracy-output.txt`.

which is not an image until something demosaics, white balances and scales it. And the tiling that the whole detection approach rests on — cutting each frame into 48 overlapping crops, submitting them, and reassembling detections back into frame coordinates — is host work by construction, as is every other thing the aircraft does: the phase machine, MAVLink routing, the mesh, survivor claim and release, the delivery excursion and the telemetry the ground station consumes.

So the three parts are a sensor, an image pipeline with a CPU, and a matrix engine. Removing any one of them does not simplify the aircraft; it stops the detection chain working.

One alternative would genuinely remove a box, and it was rejected on a measurable ground rather than a preference. Sensors now exist with an inference engine on the die — the imager runs the network itself and emits detections instead of pixels, at roughly 40 % lower power than a discrete accelerator, and it would delete the accelerator line entirely. It does not fit this mission: on-sensor engines carry a model of a few megabytes and perform a *single* inference per capture at a fixed input size. This design needs **48** inferences per frame, because a survivor occupies a few tens of pixels in a 12 MP frame and tiled inference is the only thing that makes that target detectable at all (Section IV-G). An imager that cannot tile its own output is being asked to do the one thing the architecture exists to avoid. It also removes nothing else: the host is still required for everything in the paragraph above.

That trade is worth stating because it is the obvious economy to propose, and the reason it fails is specific to small-object search rather than general.

G. Onboard processing pipeline

Fig. 4 shows what runs where. The division of labour is the part worth stating plainly, because it is what keeps the search phase deterministic: **the autopilot owns the flight, and the companion computer never touches it during the sweep.** Transects are uploaded as an ordinary AUTO mission during setup, and the aircraft would fly the entire search pattern with the companion powered off. The companion observes, detects, and takes control only for a delivery excursion — a bounded GUIDED interruption that returns the aircraft to the same AUTO mission afterwards.

That boundary matters for two reasons. A companion crash or thermal throttle costs detections, not control. And the search path is reproducible from the mission file alone, which is what makes the coverage and deconfliction results in Section VI-B checkable rather than emergent.

Detection does not gate flight. Inference runs on captured frames while the aircraft continues its transect. Losing frames to a thermal event or a slow model costs looks at a target, and the fusion requirement of twelve looks per pass carries enough redundancy to absorb some of that; it does not stall the sweep or extend the mission.

The accelerator is a substitution the inference budget has not yet been re-run against. The tiling arithmetic above was derived against a Jetson-class module, whose published throughput for comparable models spans roughly 24 to 41 frames per second. The adopted configuration instead pairs a 26-TOPS accelerator with a single-board host, which is a different architecture with a different throughput curve and a lower sustained power draw. The 24-inferences-per-second requirement is therefore stated against hardware that has not been benchmarked, and P5 benchmarking on the parts actually purchased is the gate before design freeze. Table IV carries this rather than the text implying the figure is settled.

H. Communications

Regulatory rules require the ground station to display a live camera feed from *each* aircraft. Three independent MJPEG streams would demand 4.5–6 Mbps against a link budget of approximately 2.5 Mbps. The system therefore uses H.264 encoding served through a media gateway, carrying three concurrent WebRTC streams in approximately 2.7 Mbps, with per-feed rate reduction under congestion rather than feed switching. Detection runs onboard; the downlink serves the operator, not the algorithm.

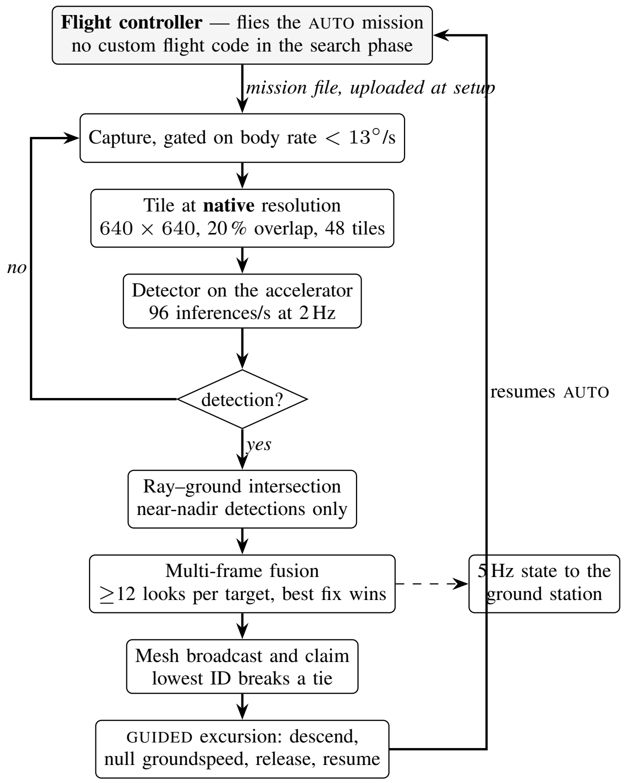

Fig. 4. Onboard processing. The autopilot flies the search pattern unaided; the companion computer detects, geolocates and claims survivors, and interrupts the mission only to deliver. Detection runs alongside the sweep and never gates it — a missed frame costs a look, not a waypoint. The capture gate suppresses turn arcs, where the ground-plane assumption is least valid and angular smear is worst. One pixel subtends 13.63 mdeg on this sensor, so at $1/1000$ s a body rate of 13.6°/s is exactly one pixel of angular blur; the gate is set at 13°/s to sit under it.

Band diversity is provided without a third radio. The 5.8 GHz mesh carries video, telemetry and coordination; a 2.4 GHz ExpressLRS link carries safety-pilot control and abort, and is independent of the autonomy stack. Both operate in delicensed bands and both comfortably cover the 600 m geofence.

The abort path is not an assumption about the link; it is a documented mode of it. ExpressLRS carries MAVLink natively from version 3.5, so the same receiver that gives the safety pilot manual control also carries the autopilot's command and telemetry stream — the ArduPilot side is a serial port set to MAVLink v2, nothing more [16]. The earlier AirPort mode achieves the same thing as a transparent serial bridge and is no longer the recommended route for MAVLink. Naming the mechanism matters here: deferring the sub-gigahertz radio (Section VIII) rests entirely on this link being able to do both jobs. A further link in the 865–867 MHz short-range-device band would add a third independent path and is the correct extension when funding allows; it is deferred rather than fitted, and Section VIII records the reasoning.

I. Payload delivery

Kits are released at 6 m above ground level, gated on a residual groundspeed below 0.3 m/s. At that release state, ballistic dispersion contributes approximately 0.32 m to total delivery error; combined with geolocation and position-hold error the root-sum-square is approximately 1.5 m with RTK active. Release altitude is chosen low enough that wind drift remains under a metre in realistic conditions, and high enough for safe clearance above a person and above debris.

Positive-lock mechanical detents on each station ensure that a power interruption cannot release a kit. This is treated as a safety-critical item and is excluded from cost reduction by policy.

J. Mission sequence

Fig. 5 shows the autonomous mission sequence. Two properties are worth drawing out. First, the launch stagger lives in the mission file as an explicit delay before each take-off command, not in operator timing — this is what converts a 1.31 m closest approach into the 64.80 m of Fig. 6. Second, every terminal branch converges on return-to-launch: link loss, low battery and operator abort share one recovery path rather than three. Third, the battery branch is *staged* rather than single-valued: the reserve threshold commands return-to-launch, and a second, lower threshold commands an immediate landing where the aircraft stands. A pack that cannot reach the pad should land under control at the point it admits this, rather than descend uncommanded partway home.

V. Sizing and Analysis

This section gives the derivations behind the design point rather than only its result. Every figure regenerates from `tools/sizing-model/rescueswarm_sizing_model.py`; the section itself is generated, so a constant cannot change in the model without changing here.

A. Constants and where they come from

Symbol	Value	Basis
ρ	1.150 kg/m^3	~30 °C at 300 m density altitude; an Indian summer field, not sea level
FM	0.60	Rotor figure of merit, mid-range of the 0.5–0.7 literature band
η_{mot}	0.82	BLDC efficiency near the hover operating point
η_{esc}	0.95	Electronic speed controller
P_{avio}	55 W	Companion computer, autopilot, GNSS, camera, radios, servos
DoD	80%	Usable depth of discharge; land at 20 % state of charge
T/W	2.0	Static thrust-to-weight, a reserve policy

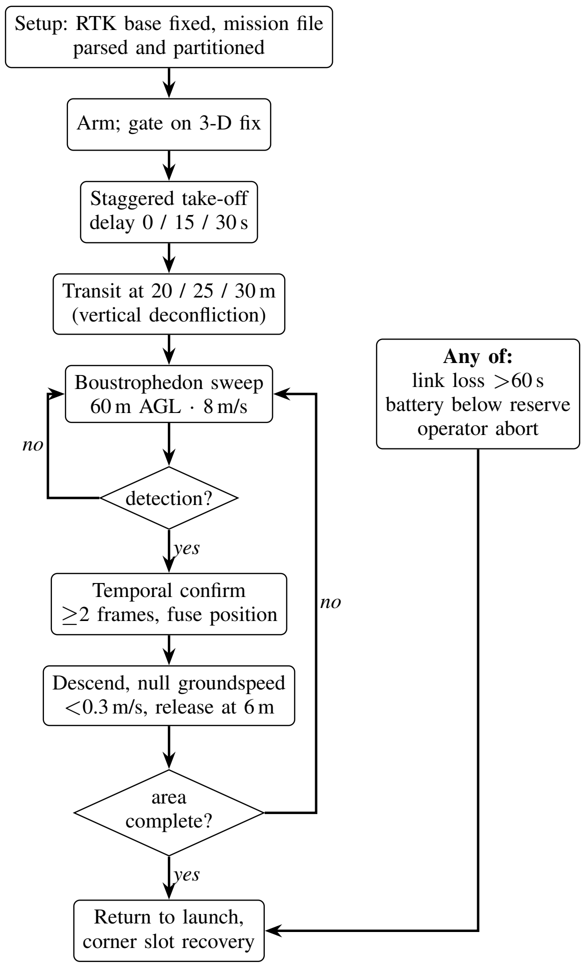

Fig. 5. Autonomous mission sequence. The take-off stagger is encoded in the mission file rather than in operator timing, and all three abnormal conditions converge on a single return-to-launch path.

Two of these are deliberately unflattering. Air density is taken at a hot-day altitude rather than sea level, which *increases* the power required; and the figure of merit is mid-band rather than optimistic. Sizing errors that favour the design are the ones that are discovered on the flight line.

B. Hover power, from momentum theory

The rotor disk area for 4 rotors of diameter 18 in ($D = 0.4572$ m) is

$$A = N\pi\left(\frac{D}{2}\right)^2 = 0.6567\ \mathrm{m}^2,$$
$$\text{disk loading} = \frac{m}{A} = 9.69\ \mathrm{kg/m}^2.$$

Momentum theory gives the induced power required to generate thrust T across that disk [17]. Dividing by the figure of merit FM — the ratio of ideal induced power to actual shaft power, which absorbs profile drag and non-uniform inflow — converts the ideal figure to shaft power:

$$P_{\text{shaft}} = \frac{T^{3/2}}{\mathrm{FM}\sqrt{2\rho A}} = \frac{(62.4)^{3/2}}{0.60\sqrt{2(1.150)(0.6567)}} = 668\ \mathrm{W},$$

where $T = mg = 62.4$ N at the 6.36 kg maximum take-off mass. Adding the drivetrain and the avionics load,

$$P_{\text{elec}} = \frac{P_{\text{shaft}}}{\eta_{\text{mot}}\eta_{\text{esc}}} + P_{\text{avio}} = \frac{668}{0.779} + 55 = 913\ \mathrm{W},$$

which at the 21.6 V pack nominal is a hover current of $I = P/V = 42.2$ A. Power loading is 7.0 g/W.

C. Peak power and the current the pack must supply

Induced power scales as $T^{3/2}$, so at the design thrust-to-weight of 2.0 the shaft power rises by $2.0^{3/2}$:

$$P_{\text{peak}} = \frac{(2.0\,T)^{3/2}}{\mathrm{FM}\sqrt{2\rho A}\,\eta} + P_{\text{avio}} = 2481\ \mathrm{W} \implies I_{\text{peak}} = 114.8\ \mathrm{A}.$$

That current, not the energy, is what disqualifies most candidate packs. It is a continuous requirement rather than a burst rating: the aircraft must be able to hold 2.0 × weight, not pulse it. The adopted pack supplies 135 A, a margin of 18 %, and the per-motor share sets the controller rating at 29 A peak.

This is also where a smaller pack fails. At 13.5 Ah the peak is 8.51 C and hover is 3.13 C; a pack half the size would need twice the C-rate to deliver the same amperes. Capacity is how the design buys down discharge rate.

D. Propulsion

$$T_{\text{total}} = (T/W)\,mg = 125\ \mathrm{N} = 12.72\ \mathrm{kgf},$$
$$T_{\text{motor}} = 3.18\ \mathrm{kgf}.$$

Hover demands 1.59 kgf per motor, which is **50 % of maximum** — inside the 45–55 % band where propeller efficiency and thermal margin are both reasonable. Sitting far below that band means an oversized and heavy propulsion system; far above it means no authority left for attitude control in gusts.

Against the selected motor, hover is comfortable and the peak is not. The Tarot TL96020 publishes a maximum continuous current of 26.5 A. The hover draw of 10.6 A per motor is 40 % of that, with wide margin. The peak at the design thrust-to-weight is 28.7 A, or **108 %** — above continuous rating. That is acceptable only because $T/W = 2.0$ is a transient authority reserve rather than a flight condition: the aircraft hovers at half throttle and reaches this current only in a gust recovery or an aggressive manoeuvre, for seconds. It is recorded here because it is the tightest margin in the propulsion chain, and because the P2 thrust stand measures current alongside thrust and will settle whether the motor reaches 3.18 kgf before that rating rather than after it. The datasheet also gives a motor mass of 168 g against the 160 g the sizing loop assumes, a difference of +32 g per aircraft that the P5 weigh-in resolves.

E. The coupled mass and energy solve

Pack mass, take-off mass and mission energy are mutually dependent: a heavier pack raises hover power, which raises the energy required, which raises pack mass. The model iterates to a fixed point rather than assuming one.

The reserve policy is what closes it. The pack must carry the nominal mission, one complete re-sweep of the search area, and four minutes of loiter, inside the usable window:

$$\underbrace{103.4}_{\text{nominal}} + \underbrace{30.9}_{\text{re-sweep}} + \underbrace{60.8}_{\text{loiter}} = 195.2 \text{ Wh usable},$$

against 233.3 Wh available at 80% depth of discharge — a margin of **19 %**. Expressed independently of chemistry, any candidate pack must therefore provide **244 Wh** nameplate at **6S** and **114.8 A continuous**.

The adopted implementation is 18 cells in 6S3P: 292 Wh, 13.5 Ah, 1449 g, 21.6 V nominal (18–25.2 V). Hover endurance is 15.3 min against a 7.6 min design mission consuming 103.4 Wh.

At 6.36 kg per aircraft the fleet is **19.08 kg** against the 25.0 kg regulatory cap, leaving 24 % margin.

F. Camera and optics

Everything optical follows from three numbers: pixel count, pixel pitch and focal length. Active area is count times pitch,

$$w = 4056 \times 1.55\,\mu\text{m} = 6.287 \text{ mm},$$
$$h = 4.712 \text{ mm}, \qquad d = 7.86 \text{ mm},$$

and the fields of view are exact arctangents, with no small-angle approximation:

$$\theta = 2\arctan\left(\frac{s}{2f}\right)$$
$$\Rightarrow\ \text{HFOV} = 55.3^\circ,\ \text{VFOV} = 42.9^\circ,\ \text{DFOV} = 66.4^\circ.$$

Sensor and ground are similar triangles about the lens, so ground sample distance at altitude H is

$$\text{GSD} = \frac{p\,H}{f} = \frac{1.55\,\mu\text{m} \times 40\,\text{m}}{6\,\text{mm}} = 1.03 \text{ cm/px},$$

giving a 41.9×31.4 m footprint and a 1.7 m supine adult subtending **165 px** at 40 m. One pixel subtends 14.80 mdeg.

Focus. This is not a design problem, and the arithmetic says so. The hyperfocal distance,

$$H_{\text{hyp}} = \frac{f^2}{Nc} + f = 4.15 \text{ m},$$

taking the circle of confusion as two pixels rather than a film-era constant, puts everything from 2.08 m to infinity in acceptable focus. The aircraft never operates below 30 m. A fixed lens set once is correct at every altitude flown, and a focus mechanism would add a moving part, a power draw and a failure mode that cannot be detected from the air.

Motion blur. This does not bind either. Holding smear under one pixel requires $t_{\text{exp}} \leq \text{GSD}/v$, which at 8 m/s is 1.29 ms — a shutter of $1/774$ s. The requirement already mandates $1/1000$ s or faster.

Rolling shutter. Small but not zero. Reading the frame out over 25 ms while the aircraft moves gives 0.20 m of top-to-bottom skew, about 19 px. Negligible for detection; *not* negligible for geolocation, where a detection's row position carries an along-track bias unless the pipeline corrects for readout row.

Temporal sampling. This is where the budget does not close. A target is in frame for the along-track footprint divided by ground speed, so at 40 m and 2 Hz it receives **7.9 looks** against the 12 that multi-frame fusion requires. Meeting that count needs

$$r \geq \frac{n\,v}{D} = 3.06 \text{ Hz},$$

Of the available levers — higher capture rate, lower ground speed, higher altitude, or a relaxed look count — raising the capture rate is the cheapest, because lowering altitude shrinks the along-track footprint faster than it shortens the pass and therefore *raises* the required rate, while raising altitude coarsens ground sampling and reducing ground speed lengthens every transect.

The cost of that lever is throughput, not energy. At 48 tiles per frame the required rate is 147 inferences per second, against the 130–160 FPS measured for the accelerator: at or beyond the limit. This is what a two-stage gate would buy back. A gate answering only whether a tile is homogeneous open water is a texture discrimination rather than a person detection, so it survives the downsampling that Section IV-E shows would destroy a person; at 96×96 input it costs roughly 44 times fewer pixels than a detector pass, and discarding the open water would leave of order 22 full-inference equivalents. On that argument the gate belongs on the accelerator already carried, not on an additional processor: the saving is in inferences, not in watts, and the aircraft's compute power is under one per cent of its hover power.

G. Payload release ballistics

A kit of 200 g tumbling with drag coefficient 1.05 and reference area 117 cm^2 has ballistic coefficient $\beta = m/(C_d A) = 16.3$ kg/m^2 and terminal velocity

$$v_{\text{term}} = \sqrt{\frac{2mg}{\rho C_d A}} = 16.2 \text{ m/s}.$$

Integrating the fall with quadratic drag from a 6 m release gives 1.15 s to impact at 9.7 m/s, with a crosswind drift of 3.24 m at 3 m/s and 6.13 m at 6 m/s.

The governing sensitivity is release velocity, not release height. A residual ground speed of 0.5 m/s displaces the impact point by about 0.53 m, whereas a full metre of altitude error contributes under 4 cm. This is the whole argument for hover-and-drop: a multirotor can null its ground speed before release and a fixed-wing cannot.

H. Structure, geometry and control authority

Tip clearance of 30 mm between adjacent 18 in rotors sets a diagonal wheelbase of 689 mm and an arm length of 0.345 m. At the design thrust-to-weight the root bending moment is

$$\mathcal{M} = T_{\text{motor}}\,L = 10.7 \text{ N m}.$$

For a 25×23 mm carbon tube, $I = \pi(D_o^4 - D_i^4)/64 = 5438$ mm^4, so the bending stress is $\sigma = \mathcal{M}c/I = 25$ MPa against roughly 600 MPa for unidirectional carbon — a safety factor of about **24**.

Bending is therefore not the structural driver, and the design should not be optimised as though it were. Joint and clamp design, and vibration fatigue at the arm root, are what actually govern; a tube chosen to satisfy the bending case by a factor of 24 is being chosen for stiffness and handling, not strength.

Releasing one kit from a magazine offset from the centreline shifts the centre of gravity by under 2 mm, requiring a few tens of grams of differential thrust to trim — trivial in magnitude, but a *step* change that excites the attitude loop. The mitigation is to mount the magazine on the centre-of-gravity axis and hold position briefly after release rather than to trim harder.

I. Radio link budget

Free-space path loss at the 600 m design slant range is $\mathrm{FSPL} = 20\log_{10} d + 20\log_{10} f + 32.44$:

Link	FSPL	Role
5.8 GHz mesh	103.3 dB	Data and video; margin is thin and degrades first
2.4 GHz mesh	95.6 dB	Fallback
868 MHz safety	86.8 dB	Command of last resort; large margin by design

The ordering is the design intent: the highest-rate link is the one permitted to fail, and the lowest-rate link is the one that must not. Video degrades before telemetry, and telemetry before command.

J. What this section does not establish

Every figure above is *modelled*. The thrust figures assume a motor that publishes no thrust curve, and the detection figures assume a recall that has not been measured. The sizing is internally consistent and its arithmetic is checkable; that is a different claim from saying the aircraft has been shown to do these things, and this document does not make the second claim anywhere.

VI. Methodology and Work Plan

The programme is organised into ten phases. Phases P1–P5 are analysis, procurement and integration; P6–P10 are progressively more demanding flight test. The eight-week flight-test window spanned by P6–P10 is the binding schedule constraint on the entire programme, and the work plan is arranged to protect it.

TABLE III
Programme phases

Ph.	Title	Principal exit criterion
P1	Requirements and sizing	Design point frozen; all models regenerate
P2	Ground segment	Single-operator interface demonstrated against three simulated aircraft
P3	Autonomy	Deterministic plan generation; deconfliction verified in simulation
P4	Fault handling	Link-loss and low-battery behaviour verified by fault injection
P5	Procurement and build	Three airframes assembled and weighed; thrust stand complete
P6	First flight	Hover, trim, vibration and thermal characterisation
P7	Perception field trials	Detection recall measured on real imagery against fabricated targets on RTK-surveyed ground
P8	Delivery trials	Delivery CEP measured against a surveyed target mat
P9	Full mission rehearsal	End-to-end mission with all three aircraft, no operator intervention
P10	Setup and contingency	Setup drill timed over repeated runs; margin recovered

A. Onboard health monitoring

Two further inference tasks run on board, and neither is in the perception path. Both are one-dimensional signal problems small enough for microcontroller-class execution, so they consume none of the accelerator budget that Section IV-E shows to be tight.

Propulsion anomaly detection. Per-motor current and airframe vibration are already logged for the characterisation in P6. The same signals identify propeller imbalance, bearing wear and mount loosening before they become in-flight failures, which matters in a programme flying eleven motors across three aircraft through an eight-week campaign. The training data is a by-product of flight testing rather than a separate collection effort.

Pack state of health. The reserve policy in Section V is a fixed 80 % depth-of-discharge rule. That rule is chemistry-correct but blind to cell ageing and pack temperature, so it is conservative early in life and optimistic late in it. A learned estimator over voltage, current and temperature under load returns usable remaining energy directly. Since Section IV-B shows the reserve, not the design mission, is what sizes the pack, tightening this estimate is the cheapest available route to either margin or endurance.

The governing rule for both is that an incorrect output costs energy or an unnecessary precautionary landing — never a missed detection. That is the condition under which a small model is admissible in this system, and it is why neither task is placed in the perception path.

B. Verification strategy

The programme's verification discipline is its most transferable contribution, and it exists because of a specific and recurring class of defect.

On four separate occasions during preparatory work, a configuration artefact was found to be correct, unit-tested, reviewed, and *not in the execution path* of the running system. In one instance a validated failsafe parameter set was never loaded by any simulation, so every test ran against stock defaults in which the low-battery action was disabled. In another, a message-routing configuration started cleanly and forwarded zero messages for weeks. Each of these reviewed clean and had passing tests around it.

The countermeasure adopted is procedural rather than technical: no capability is considered demonstrated until the real artefact has been exercised end to end and the resulting values read back off the running system. Consequently every claim in this proposal carries an explicit evidence class, and those classes are maintained mechanically rather than editorially:

- **Measured** — observed on a running system, in software or in hardware-in-the-loop simulation.
- **Modelled** — computed by a model whose inputs are themselves assumptions, and which regenerates in continuous integration.
- **Assumed** — neither measured nor modelled; carried explicitly as a risk.

A continuous-integration job enforces the rule that every published number regenerates byte-for-byte from its model, so that changing a model without committing its regenerated output fails the build. Thirteen model outputs are currently under this gate.

The same discipline now covers the cost basis. Every figure in this section is imported from a single module, which also generates the cost workbook and the budget charts; nothing is restated in two places. That was not true earlier in this programme — the workbook and this document drifted until they disagreed about the headline figure — and the fix was structural rather than editorial, because an inconsistency that can be reintroduced by hand eventually is.

VII. Preliminary Results

Substantial work has been completed before this request. we summarise it here with the evidence class of each result stated, because a claim whose basis is declared can be independently checked, whereas a claim whose basis is left implicit cannot be.

A. Measured in simulation

- **Low-battery failsafe returns the aircraft to the pad.** Verified across three simultaneous software-in-the-loop autopilots; return-to-launch triggered at 10 809 mAh of a 10 800 mAh planned consumption.
- **Three aircraft on one ground station.** A single telemetry router carried three distinct system identifiers to one ground station port with all three arriving.
- **Inter-aircraft separation at launch.** Staggered take-off delays encoded in the mission file, rather than in operator timing, increased the closest airborne approach during launch from 1.31 m to **64.80 m** (Fig. 6). Both figures are recomputed directly from the committed telemetry.
- **Separation during the sweep.** Minimum airborne separation away from the pad is **29.19 m**, with staggered transit altitudes of 20/25/30 m providing vertical deconfliction independent of the lateral plan.
- **Recovery deconfliction.** Assigning landing slots to the corners of the recovery pad rather than in a row increased slot spacing from 1.22 m to 2.61 m at no cost, raising the measured minimum approach from an overlapping 0.83 m to **2.27 m** — a clearance of 1.22 m between airframes (Fig. 7). Separation while stacked in the air over the pad improves by the same mechanism.
- **Ground station.** All eight required operator displays were verified against three simultaneous autopilots, with offline map tiles (Fig. 9). Three concurrent H.264 streams within the link budget were verified in a separate run with the media server active; the capture in Fig. 9 was taken without it, which is why the panes there read offline.

1) Pad containment has exactly zero margin: Fig. 7 shows the corner slots sitting precisely on the inset square. Nominal edge clearance is 0.523 m — one airframe radius — so no part of a stationary aircraft lies outside the box, and the geometric requirement is met by construction.

That is compliance with no margin, and it is worth stating plainly. The rules admit no part of any aircraft outside the box during launch and landing. Touchdown dispersion of ± 0.5 m, the figure this programme uses elsewhere for landing accuracy, would put up to half a metre of airframe over the edge.

The two constraints pull against each other: moving the slots inward to buy edge margin costs inter-aircraft spacing, which is precisely why they were moved to the corners. Resolving it needs a *measured* touchdown dispersion rather than an assumed one, and that is a P6 activity. It is recorded here because the geometry looks safe on the figure and is only safe on average.

B. Modelled

Endurance, hover power, mass budget, thermal margin, link budget, delivery dispersion and geolocation error are all modelled and regenerate from source. The geolocation Monte Carlo reconciles with the independent analytic budget to within one per cent, and the geotag projection has been verified self-consistent between two independent formulations to 7.8×10^{-10} m.

C. Assumed, or not yet implemented

Boresight calibration residual, achieved RTK accuracy in the field, and battery internal resistance under Indian ambient conditions are all assumed, and each is a funded activity in P5–P8. **Detection recall on real post-flood imagery is also assumed, and its field campaign is deferred rather than funded** (Section VIII); it is therefore the one assumption this programme does not currently plan to retire, and it leads the risk register accordingly.

One ground-segment function is explicitly *not* implemented and is visible as such in Fig. 9: the abort and recall controls record operator intent in the mission log but transmit nothing, because no downlink command path is yet wired. Recovery today depends on the safety pilot's transmitter. Closing this is a P4 activity, and it now runs over the 2.4 GHz ExpressLRS link rather than the deferred sub-gigahertz radio — which is what makes that link load-bearing for safety and not merely a control convenience.

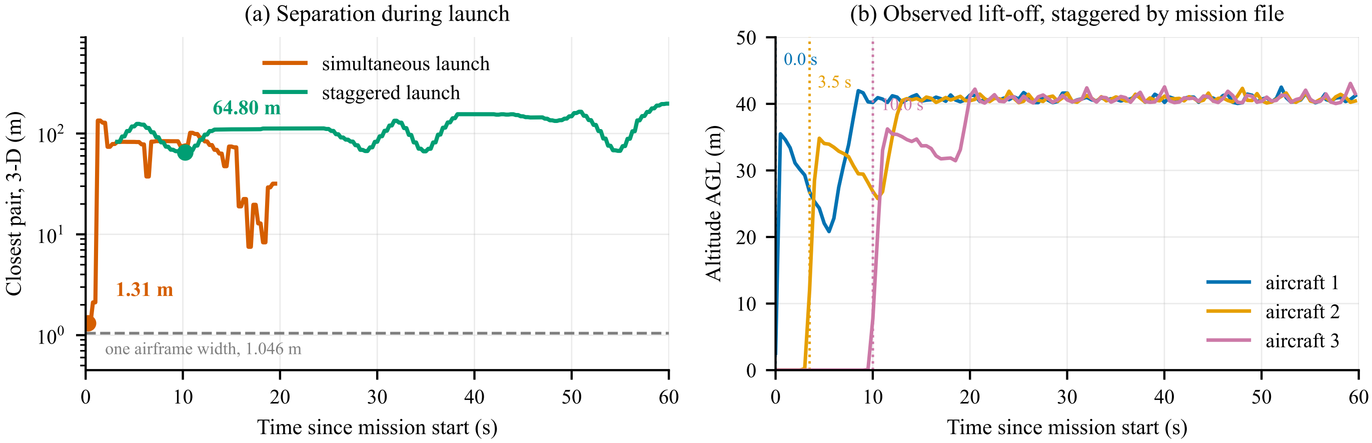

Fig. 6. Launch deconfliction, recomputed from the committed telemetry of three ArduCopter software-in-the-loop autopilots. (a) Closest pair in three dimensions, airborne aircraft only, on a logarithmic scale. Launching simultaneously brings two aircraft within 1.31 m — inside one airframe width, marked — while staggering take-off in the mission file holds the closest pair to 64.80 m. (b) The mechanism: lift-off times observed in the same recording. The mission file commands delays of 0/15/30 s; the aircraft are observed leaving the pad at 0.0/3.5/10.0 s, and **the figure plots what was observed rather than what was commanded** — see Section VII-C.

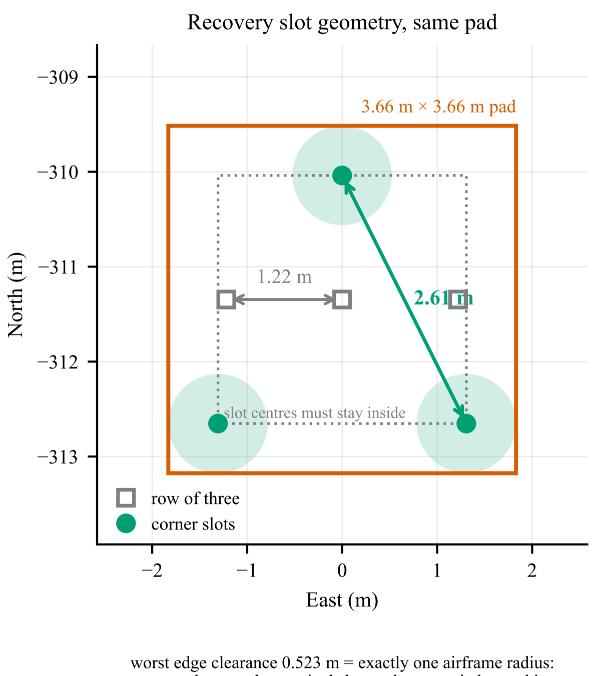

Fig. 7. Recovery slot geometry, drawn from the two committed recordings. Slot centres must stay half an airframe inside the pad edge, so they live in a 2.61 m square: a row across that square gives 1.22 m of spacing, its corners give the full 2.61 m. Shaded discs are the 1.046 m airframe footprint at the corner slots. The measured closest approach at any time rises from an overlapping 0.83 m to 2.27 m — twice the separation, on the same pad, at no cost.

D. Evidence status

Nothing in this programme has flown. Every "measured" result above was measured in simulation or in software. That rules out whole classes of logical and integration error and rules out none of the physical ones. The build phase exists principally to convert the modelled and assumed rows of this section into measured ones.

VIII. Scope of This Build

This document describes an integrated system; the funding request is held separately and is not part of it. What belongs here is not what the build costs but *what it does and does not contain*, because several arguments earlier in this document depend on it.

A. What this build deliberately does not include

Each of the following was deferred rather than judged unnecessary. Stating them is not padding: a reviewer should be able to see what the lowest-cost configuration gives up, and in two cases the omission changes what this document is entitled to claim.

The field-data campaign and the ground-truth apparatus are both deferred, and the consequence is stated plainly: ***detection recall in this document is modelled, not measured.*** There is currently no funded route to measuring it. Of everything listed here this is the omission with the largest effect on delivered capability, and it is the first item that should be reinstated if the position changes.

The second GNSS receiver is deferred, and the deferral is now forced rather than chosen. It would supply moving-baseline heading, and Section IV-D shows that substituting a magnetometer-class attitude allocation into the case-C budget costs **18 mm**. The case for it was never the error budget — it was insurance against a magnetometer sitting near a high-current bus, which the budget does not model.

The supplier has since confirmed that although both candidate receivers operate as RTK rovers, moving-baseline dual-antenna heading remains under development [15]. The capability is therefore not purchasable at the time of writing at any price, which settles the question for this build and removes it from the cost-reduction argument entirely. **P6 characterises achieved heading accuracy on the built aircraft; if it**

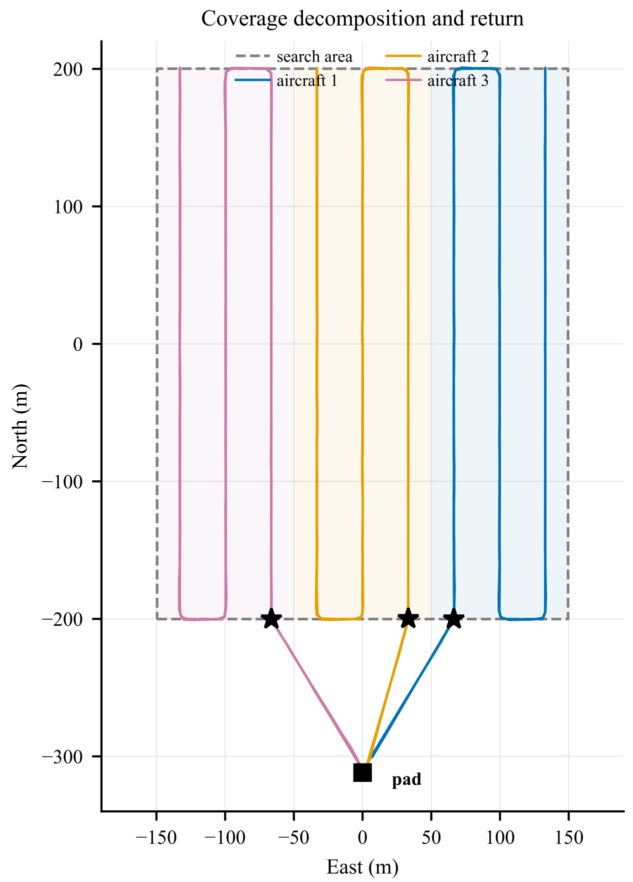

Fig. 8. Coverage decomposition and return geometry, drawn from the committed telemetry. Each aircraft is assigned an equal-area strip (shaded) and flies a boustrophedon pattern within it; each strip is swept twice, on opposing headings, so that boresight error cancels rather than accumulating. Stars mark where each sweep ends. Enumerating the four start-side and direction combinations and keeping the one that finishes nearest home brings all three aircraft to within 117–130 m of the pad, against 516 m and 540 m for two of three under the previous index-keyed rule — at identical path length, and on the lowest state of charge of the flight.

exceeds roughly 1°, a heading source must be reinstated — by then either through firmware maturity at this supplier or through an imported dual-antenna receiver.

The sub-gigahertz safety radio is deferred, and that deferral rests on a configuration rather than on a judgement. The primary link must run in a mode that carries both control and MAVLink telemetry on one link and one autopilot serial port. The alternative transparent-serial mode consumes that link, and an aircraft configured that way has telemetry and no control. **This is verified on the bench at P2; if it does not hold, the radio is not optional.**

Recovery parachutes are permitted, not required. A crash scores -50 against -10 for landing outside the zone, so the deferral is a bet of roughly forty points against the probability of a crash.

Spares cover what crashes, not what fails rarely. Propellers, motors, arms, plate stock and packs are retained. Spare autopilot, GNSS and camera are removed. **This is an accepted risk and we state it as one:** a failure of any of those three inside the flight-test window costs schedule, and the mitigation is procurement lead time rather than inventory.

Flight insurance is deferred and must be confirmed against the rulebook and the Drone Rules before any flight takes place. This is the one deferral that is not purely a performance trade.

B. One caveat on how ground truth will be measured

The ground-truth apparatus is fabricated rather than bought. Survivor targets are clothed forms of the correct size and outline; what a detector responds to is silhouette, scale and contrast against terrain, not anatomy. Two of these items are *better* fabricated: dummy kits can be made to exactly the 200 g that rule C6 fixes, and donated clothing gives wider colour and contrast variety than matched bought sets, which is precisely what detection robustness requires.

The calibration markers are surveyed using the programme's own RTK rover. That is standard practice and adequate for the purpose, but it means the ground truth against which geolocation accuracy is measured is *itself RTK-derived*. The two error sources are therefore correlated, and any measured geotag figure reported in P7–P8 will be slightly optimistic against an independently surveyed datum. Any accuracy claim arising from those phases will state this.

IX. Risk Register

We draw particular attention to the setup-time risk. It is currently supported by a model with a modest margin and no measurement, and it gates the entire mission. Timed bench drills are scheduled as the earliest hardware activity in the programme for precisely this reason.

X. Expected Outcomes and Deliverables

1) Three flight-qualified aircraft with a documented, market-verified bill of materials.
2) A single-operator ground control station demonstrated end to end against real aircraft, with no network dependency.
3) A measured detection recall figure against fabricated survivor targets on surveyed ground, replacing the current assumption. Ground truth is RTK-derived rather than independently surveyed, and the resulting figure will state that.
4) A measured delivery circular error probable against a surveyed target, replacing the current model.
5) A measured setup-to-launch time distribution over repeated drills.
6) An open, reproducible evidence base in which every published figure regenerates from its model under continuous integration.

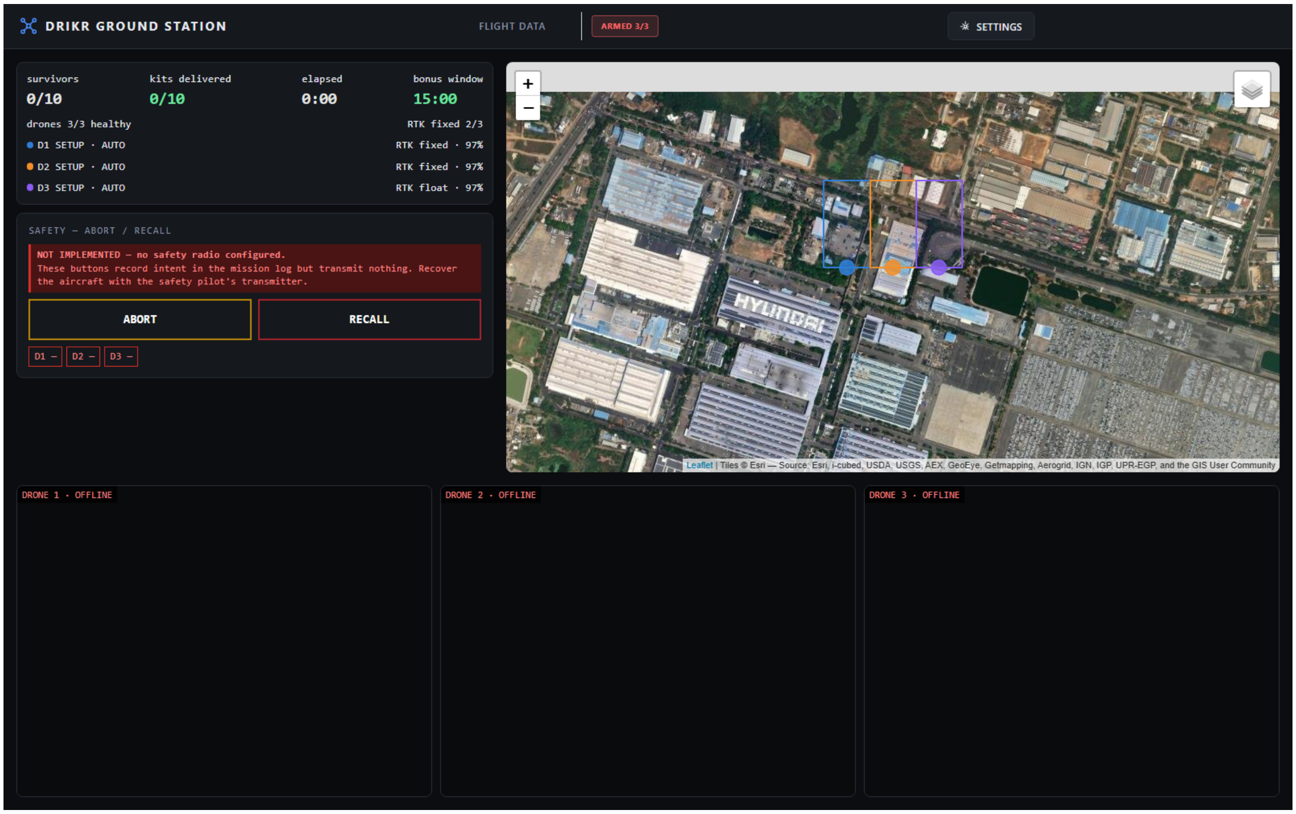

Fig. 9. Ground control station, captured unretouched from the running client and backend. **The telemetry is synthetic**: a mission driver (`scripts/sim_mission.py`) emits MAVLink and 5 Hz mission-state documents for three aircraft, so every display can be exercised before an aircraft exists. No aircraft, autopilot or radio is involved. Visible here: the three assigned search regions on a satellite basemap served from a local tile cache, so nothing depends on a network; three aircraft armed and in AUTO, two with an RTK-fixed solution and the third in float, which is the degraded case the display must distinguish; and the mission counters, where the 15:00 figure is the `SYS-35` fast-completion threshold rather than the flight-time limit. The safety panel is shown uncropped deliberately. It states that abort and recall are *not implemented*, that they record operator intent in the mission log and transmit nothing, and that recovery is by the safety pilot's transmitter — because no downlink command path is yet wired (Section VII-C). The video panes read offline because the media server is not running in this capture; no camera has yet flown.

XI. Future Work

This proposal deliberately scopes a single prototype build and nothing more. The work has a longer arc, set out here so that the boundary of the present scope is unambiguous by contrast.

A. From a prototype to a deployable system

This build is not a product. Reaching one requires, in rough order: night capability, discussed below; beyond-visual-line-of-sight approval, which is a regulatory programme rather than an engineering one; ingress protection and thermal qualification for monsoon conditions; and endurance beyond the present 15.3 minutes, which at this payload means a different airframe class rather than a better battery.

The night-sensing route is not the obvious one. Thermal imaging is the default answer, and it is genuinely valuable over rooftops, dry debris and open ground. It degrades, however, for precisely the case this system is built around. A person immersed in floodwater equilibrates toward water temperature, and the skin-to-background contrast a microbolometer depends on largely disappears; the residual signal comes from the head and shoulders alone, which is the smallest part of an already small target.

Near-infrared is the better-matched extension. Water absorbs strongly beyond roughly 900 nm while skin remains comparatively reflective, so a person presents as a bright object against dark water — the same contrast mechanism that makes NIR standard for water-body segmentation in remote sensing. It is also an order of magnitude cheaper than a thermal core, since it requires a camera with the infrared-cut filter removed rather than a new sensor class, and it can be evaluated against public imagery before any hardware is bought. The two are complementary rather than alternative: thermal for survivors on dry ground at night, near-infrared for survivors in water.

B. Where the compute cliff falls

Section IV-E establishes that this task cannot be met by shrinking the input, because person-pixels and coverage rate trade against one another for a fixed sensor: a smaller detector input demands either a lower altitude or a longer lens, and both narrow the swath. What that argument does not establish is *where* the boundary lies.

TABLE IV
PRINCIPAL RISKS AND MITIGATIONS

Risk	Sev.	Mitigation
Detection recall below target on real imagery	High	**Raised by deferral of the field data campaign.** Public aerial-person datasets carry the model; tiling and fusion parameters remain tunable without hardware change
Inference and software video encode contend on one board	High	The replacement compute has no hardware H.264 encoder. Benchmark both loads together before design freeze; fallback is a reduced tile count on a centre crop
Moving-baseline heading unavailable on the selected receiver	Medium	**Downgraded from High.** The supplier has confirmed RTK rover operation on both candidate receivers, closing the primary concern, but reports moving-baseline dual-antenna heading as still under development [15]. Heading therefore rests on the magnetometer, which costs 18 mm in the case-C budget; P6 measures achieved heading accuracy and reinstates a dedicated source if it exceeds 1°
Flight-test window compressed by procurement delay	High	**Raised by deferral of the training airframe and of spares.** Lead time is now the only mitigation, so long-lead items are ordered in tranche 1
Recovery parachute sizing and supply	Medium	No domestic supplier identified; import lead time and correct mass rating are open procurement actions
Setup time exceeds allowance	Medium	Currently modelled, not measured; timed drills are the highest-priority early test
Airframe overhangs the pad edge on touchdown	Medium	Nominal clearance is exactly one airframe radius, so dispersion has no margin. Measure touchdown dispersion in P6 before accepting the slot geometry
Motor thrust below the datasheet claim	High	Selected motors publish no thrust curve, and the stand cannot precede the motors. Thrust is measured in P5 on arrival; a shortfall then forces a propulsion change with the flight-test window already open
Airframe loss during test	High	**Raised by deferral of spares.** Structure set and consumables retained; a second loss in the same week is uncovered
Flight insurance may be mandatory	Medium	Deferred at team direction. To be confirmed against the rulebook and DGCA rules before any flight

Measuring it is a contribution in its own right. The same tiled task, on the same imagery, can be run at three compute tiers — the 26 TOPS accelerator specified here, its 13 TOPS variant, and a microcontroller-class part — and the altitude, tile count and inference rate at which each ceases to meet the recall requirement recorded. The result is a curve of achievable coverage rate against inference power, which is the quantity a designer actually needs and which is not, to our knowledge, published for aerial person detection.

The application is planetary aviation, where available power is a hard constraint rather than a budget line. *Ingenuity* flew 72 sorties on Mars from a 1.8 kg airframe in an atmosphere below one per cent of Earth's density, and *Dragonfly* is manifested for Titan in 2028; in both cases command latency is measured in minutes, so every perception decision is taken on board. Microcontroller-class inference is routinely proposed for such missions. The measurement above would establish what target size that choice actually forfeits.

C. Cost at production volume

The parts basis for this build is single-unit retail for a build of three. The marginal cost of an additional aircraft is already substantially below the per-aircraft figure quoted here, because the ground segment, test equipment and spares are one-off. At production volume the relevant questions are different ones — tooling amortisation, negotiated component pricing, and assembly labour — and none of them can be answered honestly from a three-aircraft bill of materials. We decline to extrapolate them.

D. A route to commercialisation

The natural customers are state disaster management authorities and the national and state disaster response forces. Establishing whether that is a viable business requires three things this programme does not attempt: the size and refresh rate of the addressable fleet, what a comparable capability is currently procured for, and the operating cost per flight hour of the boat or helicopter search it would displace. Those are questions for published tender data and operator interviews, not for a bill of materials. They are named here as open rather than filled with plausible figures.

What this programme does produce toward that end is the seventh deliverable in Section X: a component-level map of where Indian UAV supply chain capability exists and where it does not. That artefact outlives the build regardless of what follows it.

XII. CONCLUSION

We propose an autonomous three-aircraft system that addresses the localisation bottleneck in post-flood search and rescue, engineered under a verification discipline that distinguishes what has been demonstrated from what has merely been designed. Where a part was deferred it was deferred on cost, not on capability, and Section VIII states what each omission costs the system.

The system's design point, error budgets and cost basis are established and reproducible. Its coordination and failsafe behaviour have been demonstrated in hardware-in-the-loop simulation across three simultaneous autopilots. What remains is to build the aircraft, fly them, and replace the modelled and assumed quantities in Section VI-B with measured ones.

The resources this requires are documented separately. What this document undertakes is narrower: that every number in it is checkable, and that those not yet measured say so.

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